

does not provide any guarantee of optimality, however. In fact, existing empirical results demonstrate that the median heuristic performs poorly (i.e. it leads to low test power) when differences between the two distributions occur at a lengthscale that differs sufficiently from the median inter-sample distance (Gretton et al., 2012b, Figure 1). Ramdas et al. (2015) and Reddi et al. (2015) show that the median heuristic scales as the square root of the dimension, and that using a bandwidth of the higher order with respect to the dimension generally leads to higher power. Another approach is to split the data and learn a good kernel choice on data held out for this purpose (e.g. Gretton et al., 2012b; Liu et al., 2020), however, the resultant reduction in data for testing can reduce overall test power at smaller sample sizes.

Our contributions. Having motivated the problem, we summarize our contributions. We first address the case where the “smoothness parameter” s of the task is known: that is, the distributions being tested have densities in $\mathbb{R}^d$ whose difference lies in a Sobolev ball $\mathcal{S}_d^s(R)$ with smoothness parameter s and radius R . For this setting, we construct a single MMD test that is optimal in the minimax sense over $\mathcal{S}_d^s(R)$, for a specific choice of bandwidths which depend on s .

In practice, s is unknown, and our test must be adaptive to it. We therefore construct a test which is adaptive to s in the minimax sense, by aggregating across tests with different bandwidths, and rejecting the null hypothesis if any individual test (with appropriately corrected level) rejects it. We refer to our proposed MMD aggregated test as MMDAgg. By upper bounding the uniform separation rate of testing of MMDAgg, we prove that it is optimal (up to an iterated logarithmic term) over the Sobolev ball $\mathcal{S}_d^s(R)$ for any $s > 0$ and $R > 0$.

For the practical deployment of our test, we require numerical procedures for computing the test thresholds. We may obtain the threshold for a test of level α using either permutations or a wild bootstrap to estimate the $(1-\alpha)$ -quantile of the test statistic distribution under the null hypothesis. We prove that our theoretical guarantees still hold under both test threshold estimation procedures. In the process of establishing these results, we demonstrate the equivalence between using a wild bootstrap and using a restricted set of permutations, which is of independent interest. Through the use of either permutations or a wild bootstrap, we can theoretically guarantee that our proposed single MMD test and MMDAgg test both have non-asymptotic level, which differs from the original MMD test of Gretton et al. (2012a). We believe that this non-asymptotic property of our tests contributes to their real-world applications. Indeed, in practice, settings in which the sample sizes are fixed and may not be assumed to be asymptotically large are very common (e.g. medical data, seismological data, data for materials science, *etc.*). While existing tests relying on asymptotic may fail to be well-calibrated in those settings, ours are guaranteed to correctly control the test level non-asymptotically.

We stress that the implementation of MMDAgg corresponds exactly to the test for which we prove theoretical guarantees: we do not make any further approximations in our implementation, nor do we require any prior knowledge on the underlying distribution smoothness. All our theoretical results hold for any product of one-dimensional translation invariant characteristic kernels which are absolutely and square integrable. Our test is, to

the best of our knowledge, the first to be *minimax adaptive* (up to an iterated logarithmic term) for various kernels, and not only for the Gaussian kernel.

Since our approach combines multiple MMD single tests across a large collection of bandwidths, it requires no tuning and our implementation is parameter-free. Furthermore, since we consider various bandwidths simultaneously, our test is adaptive: it performs well both in cases requiring the kernel to have a small bandwidth and in those necessitating a large bandwidth. This means that the same MMDAgg test can detect both local and global differences in densities, which is not the case for a single MMD test with fixed bandwidth.

The key contributions of our paper can be summarised as follows.

- Based on either permutations or a wild bootstrap, we propose a single MMD test and theoretically prove that it has non-asymptotic level. By setting the kernel bandwidth adequately, we show that the test is minimax optimal over a Sobolev ball with known smoothness parameter.
- In order to be adaptive to this smoothness parameter (which is unknown in practice), we construct a two-sample aggregated MMD test, called MMDAgg, which does not require data splitting. We prove that MMDAgg controls the type I error non-asymptotically, and that it is optimal in the minimax sense (up to an iterated logarithmic term) for a wide range of kernels when using either permutations or a wild bootstrap to estimate the test threshold.
- In our experiments on synthetic data, on which the Sobolev smoothness assumption holds, we observe that MMDAgg obtains significantly higher power than all other state-of-the-art MMD adaptive tests considered. On real-world image data, MMDAgg almost matches the power obtained by tests leveraging the capacity of models such as neural networks to detect differences in image distributions. The power of MMDAgg is retained even when a large collection of kernels is considered (up to 12000 kernels in the experiments). Experimentally, no cost in power is incurred for aggregating more kernels, the overall power appears to match the highest power achieved with a single kernel while correctly controlling the test level. As such, in practice, the user can consider as many kernels as computationally feasible.
- We provide a user-friendly parameter-free implementation of MMDAgg, both in Jax and in Numpy, available at <https://github.com/antoninschrab/mmdagg-paper>. This repository also contains code for the reproducibility of our experiments.

Related Works. We present an overview of works related to ours, more details are provided in Section 4. Our non-asymptotic aggregated test, which is minimax adaptive over the Sobolev balls $\{\mathcal{S}_d^s(R) : s > 0, R > 0\}$, originates from the works of Fromont et al. (2012, 2013) and Albert et al. (2022).

Fromont et al. (2012, 2013) consider the two-sample problem with sample sizes following independent Poisson processes. In this framework, they construct an aggregated test using a different kernel-based estimator with a wild bootstrap. In the multivariate setting, with some condition on the kernel in the Fourier domain, they show minimax adaptivity of their test over Sobolev and anisotropic Nikol’skii-Besov balls. Albert et al. (2022) construct an aggregated independence test using Gaussian kernels. The theoretical guarantees for the

single MMD tests using a permutation-based threshold are related to the result of Kim et al. (2022, Proposition 8.4), also for Gaussian kernels. Besides treating the adaptive two-sample case, rather than the independence case considered by Albert et al. (2022), the present work builds on these earlier results in two important ways. First, the optimality of our aggregated test is not restricted to the use of a specific kernel; it holds more generally for many popular choices of kernels. Second, our theoretical guarantees for this adaptive test are proved to hold even under practical choices for the test threshold: namely the permutation and wild bootstrap approaches.

In this paper, we propose quadratic-time aggregated tests for two-sample testing. This present work, along with those of Albert et al. (2022) and Schrab et al. (2022a) on independence and goodness-of-fit testing, respectively, have been the basis of the later work of Schrab et al. (2022b) who construct efficient (linear-time) variants of these three aggregated tests using incomplete U -statistics, and quantify the trade-off between computational efficiency and test power (in terms of uniform separation rate over Sobolev balls).

Outline. The paper is organised as follows. In Section 2, we formalize the two-sample problem, review the theory of statistical hypothesis testing, and recall the definition of the Maximum Mean Discrepancy. In Section 3, we construct the MMD single and aggregated (MMDAgg) two-sample tests, provide pseudocode for the latter, and derive theoretical guarantees for both. Having introduced the required terminology, we then discuss how our results relate to other works in Section 4. We run various experiments in Section 5 to evaluate how well MMDAgg performs compared to alternative state-of-the-art MMD adaptive tests. The paper closes with discussions and perspectives in Section 6. Proofs, additional discussions, and further experimental results are provided in the Appendices.

2 Background

First, we formalise the two-sample problem in mathematical terms.

Two-sample problem. Given independent samples $\mathbb{X}_m := (X_i)_{1 \leq i \leq m}$ and $\mathbb{Y}_n := (Y_j)_{1 \leq j \leq n}$, consisting of i.i.d. random variables with respective probability density functions p and q on $\mathbb{R}^d$ with respect to the Lebesgue measure, can we decide whether $p \neq q$ holds?

To tackle this problem, we work in the non-asymptotic framework and construct two nonparametric hypothesis tests in Section 3: a single MMD test for fixed kernel/bandwidth, and MMDAgg which aggregates multiple kernels/bandwidths. In Section 2.1, we first introduce the required notions about hypothesis testing. We then recall the definition of the Maximum Mean Discrepancy and present two estimators for it in Section 2.2.

2.1 Hypothesis testing

We use the convention that $\mathbb{P}_{p \times q}$ denotes the probability with respect to $X_1, \dots, X_m \stackrel{\text{iid}}{\sim} p$ and $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} q$ all independent of each other. If given more random variables, say $Z_1, \dots, Z_t \stackrel{\text{iid}}{\sim} r$ for some probability density or mass function r , we use the notation $\mathbb{P}_{p \times q \times r}$. We follow similar conventions for expectations and variances.

We address this two-sample problem by testing the null hypothesis $\mathcal{H}_0: p = q$ against the alternative hypothesis $\mathcal{H}_a: p \neq q$. Given a *test* Δ which is a function of $\mathbb{X}_m$ and $\mathbb{Y}_n$,